

"OptiCharge" EV Charging Scheduler

Problem statement

"OptiCharge" is a company operating a growing network of electric vehicle (EV) charging stations. One of its facilities has a parking area with a limited number of charging stations, which are shared by employees and visitors throughout the day.

Drivers typically provide information about their vehicle when arriving at the facility, including their expected departure time, current battery level, and desired battery level. The charging stations have limited power capacity, and the facility cannot charge all connected vehicles at their maximum charging rate simultaneously.

Currently, charging is scheduled using a simple first-come-first-served strategy. This approach can lead to inefficient use of the available charging capacity: some vehicles may finish charging much earlier than necessary, while other vehicles may not receive enough charge before their planned departure time. Furthermore, electricity prices can vary throughout the day, making it potentially cheaper to charge some vehicles at different times.

Our goal is to automate this process. We need an intelligent software tool that can take a list of charging requests and produce an efficient charging schedule that makes the best use of the available charging infrastructure.

Project objective

Your team's mission is to develop a proof-of-concept application, codenamed "OptiCharge," that algorithmically determines when and how each electric vehicle should be charged, while satisfying the requirements of the vehicles and the limitations of the charging infrastructure.

The primary goal is to maximize the amount of requested energy delivered to vehicles before their departure. Among schedules that deliver a similar amount of energy, the algorithm should minimize the total cost of electricity.

Core needs

At the very least, the application must be able to:

1. **Read charging request data** - Read a list of charging requests from a specified file. The file will contain information about each vehicle, including:
 - arrival time;
 - expected departure time;
 - current battery level;
 - battery capacity;
 - desired battery level.
2. **Utilize available charging infrastructure** - Take into account the characteristics and limitations of the available charging stations. The charging infrastructure will be described in the input data and may include:
 - a fixed number of charging stations;
 - maximum charging power of each station;
 - maximum total power available to the facility;
 - electricity price for different time periods.

A vehicle can only be charged while it is present at the facility and the charging schedule must not exceed the available charging capacity. Unless otherwise specified, all charging stations are compatible with all vehicles.

3. **Calculate an efficient charging schedule** - Implement an algorithm that determines when each vehicle should be charged and at which charging station. The generated schedule should satisfy the charging requirements and infrastructure constraints whenever possible.

The primary optimization goals are:

- maximize the amount of requested energy delivered to vehicles before their departure;
- minimize the total cost of electricity used for charging.

If it is not possible to fully satisfy all charging requests, the application should produce the best schedule it can and clearly indicate which requests could not be fully satisfied.

4. **Output the charging schedule** - The program must be able to produce two outputs:
 - A visual representation of the charging schedule, indicating vehicles, charging stations, start/end times, charging power and electricity costs.
 - A text-based output file containing the exact tasks for operator to be carried out throughout the day in the facility - which chargers to connect (disconnect) to which vehicles and at which times.